

Audio Programmer & Software Engineer specializing in game audio, audio systems, and interactive technologies. Expertise in developing & integrating audio tools, game engines, middlewares, and other innovative solutions for AAA game projects. Proven ability in research, technical sound design, and cross-disciplinary collaboration. Holder of patents & [publications](#) with extensive experience in signal processing, machine learning, and HCI (Human-Computer Interaction).

TECHNICAL SKILLS

- **Game Engine & Middleware:** Unreal Engine, Unity, Wwise, FMOD
- **DAW:** Pro Tools, Ableton Live
- **SDKs & Frameworks:** JUCE, Google Resonance Audio, Kinect
- **Programming:** C++, C#, Python, Java, MATLAB, Max/MSP, JavaScript, Ruby
- **Platform:** Mobile (Android, iOS), PC, Web
- **Database:** MySQL, Postgres, NoSQL, Spatial-DBs, MapReduce
- **DSP/AI tool:** NumPy, Librosa, Scipy, PyTorch, NLTK, Pandas, TensorFlow
- **Others:** VR / AR, Agile methodology, Git, Technical Sound Design, Recording & Mixing, Interdisciplinary research

EDUCATION

- **MSc. in Computer Science:** University of Southern California
- **Honors BEng. in Computer Science & Engineering:** Ho Chi Minh City University of Technology (HCMUT), Vietnam
- Other Certificates/Courses
 - **Virtual Acoustics in Immersive Audio Workshop:** Stanford University CCRMA
 - **Deep Learning for Music Information Retrieval Workshop:** Stanford University CCRMA
 - **Certificate in Audio Production:** University of Washington, Seattle
 - **Ableton Live Techniques: Synthesis and DSP:** Berklee Online
 - **Game Audio:** School of Video Game Audio

PROFESSIONAL EXPERIENCE

AUDIO PROGRAMMER / LEAD TECHNICAL SOUND DESIGNER, *Formosa Interactive - Seattle, WA* 2023 – Present

- Architecting, developing and optimizing audio systems & integration for game engines & middleware. Code reviewing & bug fixing.
- Collaborating with 10+ cross-functional teams, including audio directors, sound designers, QA/Test, developers and production.
- Notable Projects:
 - Bungie's [Destiny 2](#) (released - AAA): Audio integration/implementation for The Final Shape & Season of the Wish (2023-2024)
 - [Unannounced Project - AAA]: Lead audio programming role (2024-2025)
 - Moon Studios' [No Rest for the Wicked](#): Audio programming & optimization (2025-present)

AUDIO PROGRAMMER, *Games for Love 501©(3) – Seattle, WA* 2021 – 2022

- Integrated audio assets into VR game Channel Surfing+ using Unity, C#, FMOD. Details on [link](#).
- Coordinated with 1 audio lead, 1 sound designer & 2 lead developers to ensure audio assets accuracy.

SOFTWARE DEVELOPER, *GravelVoice – Seattle, WA* 2020 – 2021

- Developed & published [mobile app](#) playing spatial music (binaural, ambisonics) using Google Resonance Audio, Oculus Audio Spatializer, Unity, C#. Collaborated with 1 music producer/audio engineer.

INDEPENDENT RESEARCHER, *Seattle, WA* 2020 – Present

- Program committee member & reviewed 10+ manuscripts for top-tier journals & conferences in the field of human-computer interaction, signal processing, audio, data mining & artificial intelligence: [CHI](#), [CSCW](#), [ICMI](#), [AM](#), etc.

TECHNICAL AUDIO INTERN, *Women's Audio Mission (WAM) – San Francisco, CA* 01/2021 – 05/2021

- Taught Python and visual coding for music production to 10+ students in [Girls on The Mic](#) program.
- Produced video/audio content for WAM's social content and events using Pro Tools, RX, After Effects.

RESEARCH FELLOW, *IBM Research AI – Yorktown Heights, NY* 05/2017 – 09/2017

- Developed Simpler Voice – a [patented](#) NLP system for low-literate adults by decoding complex texts into simple messages, visual, audio elements. Published 4 patents & Presented at NeurIPS2017, D4GX2018 conferences. Featured on [Mashable](#), [ACM](#), [IBM](#).

RESEARCH ASSISTANT, *USC Integrated Media Systems Center – Los Angeles, CA* 08/2014 – 12/2017

- Developed & deployed a secured in-hospital Kinect system & at-home client-server system collecting 100+GB data of 50+ cancer patients at 4 hospitals/clinics. Presented at AMIA2016, ICDM2017 conferences. Featuring on [Forbes](#), [NBC4News](#), [ScienceDaily](#).
- System demo at the White House as part of the [Former Vice President / President Joe Biden's Cancer Moonshot initiative](#).
- Published 4 papers. Collaborated with a team of 20+ Oncologists, Physicians, Biomedical scientists.

HONORS & AWARDS

2025. Stanford CCRMA scholarship - Virtual Acoustics in Immersive Audio
2024. Stanford CCRMA scholarship - Deep Learning for Music Information Retrieval
2021. Game Audio Diversity Alliance scholarship
2020. SoundGirls' Leslie Ann Jones scholarship in Honor of Ethel Gabriel
2014-2017. Vietnam Education Foundation fellowship (Top 40 national STEM students)
2017. IBM Social Good fellowship
2016. USC Grace Hopper scholarship
2009-2014. HCMUT Dean's Honor scholarships

OTHER PROFESSIONAL & COMMUNITY SERVICES

Member: AES, IEEE, SoundGirls, Grammy U
Volunteer: WAM, The Vera Project (Live Audio), Grace Hopper Conference, ICDM
Mentor: Mentored 10+ undergrad and high school students in audio and computer science research projects